Transportation schedule :

$S_1 \rightarrow D_1, S_1 \rightarrow D_2, S_2 \rightarrow D_2, S_2 \rightarrow D_4, S_3 \rightarrow D_3, S_3 \rightarrow D_4$

This initial transportation cost

$= (200 \times 11) + (50 \times 13) + (175 \times 18) +$

$(125 \times 10) + (275 \times 13) + (125 \times 10)$

= ₹ 12,075

### Example 10.6

Obtain an initial basic feasible solution to the following transportation problem using Vogel's approximation method.

Ware houses — Stores

| | I | II | III | IV | Availability ($a_i$) |
|---|---|---|---|---|---|
| A | 5 | 1 | 3 | 3 | 34 |
| B | 3 | 3 | 5 | 4 | 15 |
| C | 6 | 4 | 4 | 3 | 12 |
| D | 4 | 1 | 4 | 5 | 19 |
| Requirement ($b_j$) | 21 | 25 | 17 | 17 | |

***Solution:***

Here $\sum a_i = \sum b_j = 80$

(i.e) Total Availability =Total Requirement

∴ The given problem is balanced transportation problem.

Hence there exists a feasible solution to the given problem.

**First Allocation:**

| | I | II | III | IV | $a_i$ | Penalty |
|---|---|---|---|---|---|---|
| A | 5 | 1 | 3 | 3 | 34 | 2 |
| B | 3 | 3 | 5 | 4 | 15 | 1 |
| C | 6 | 4 | 4 | 3 | 12 | 1 |
| D | 4 | (19) 1 | 4 | 5 | 19/0 | (3) |
| $b_j$ | 21 | 25/6 | 17 | 17 | | |
| Penalty | 1 | 2 | 1 | 1 | | |

**Second Allocation:**

| | I | II | III | IV | $a_i$ | Penalty |
|---|---|---|---|---|---|---|
| A | 5 | (6) 1 | 3 | 3 | 34/28 | (2) |
| B | 3 | 3 | 5 | 4 | 15 | 1 |
| C | 6 | 4 | 4 | 3 | 12 | 1 |
| $b_j$ | 21 | 6/0 | 17 | 17 | | |
| Penalty | 2 | 2 | 1 | 1 | | |

**Third Allocation:**

| | I | III | IV | $a_i$ | Penalty |
|---|---|---|---|---|---|
| A | 5 | (17) 3 | 3 | 28/11 | (2) |
| B | 3 | 5 | 4 | 15 | 1 |
| C | 6 | 4 | 3 | 12 | 1 |
| $b_j$ | 21 | 17/0 | 17 | | |
| Penalty | 2 | 1 | 1 | | |

**Fourth Allocation:**

| | I | IV | $a_i$ | Penalty |
|---|---|---|---|---|
| A | 5 | 3 | 11 | 2 |
| B | 3 | 4 | 15 | 1 |
| C | 6 | (12) 3 | 12/0 | (3) |
| $b_j$ | 21 | 17/5 | | |
| Penalty | 2 | 1 | | |

**Fifth Allocation:**

| | I | IV | $a_i$ | Penalty |
|---|---|---|---|---|
| A | 5 | (5) 3 | 11/6 | (2) |
| B | 3 | 4 | 15 | 1 |
| $b_j$ | 21 | 5/0 | | |
| Penalty | 2 | 1 | | |

**Sixth Allocation:**

| | I | $a_i$ | Penalty |
|---|---|---|---|
| A | (6) 5 | 6/0 | — |
| B | (15) 3 | 15/0 | — |
| | 21/0 | | |
| Penalty | (2) | | |



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